Graduation Project Proposal

An Interactive Multi-Agent Courtroom Simulation System for Law Students

Department of Computer Science — Artificial Intelligence Track
Alexandria National University
Team size: 8 members

1. Introduction & Problem Statement

Legal education traditionally relies on moot court exercises and role-play to train students in courtroom procedure and advocacy. These methods are effective but resource-intensive, difficult to scale, and dependent on the availability of faculty, practicing lawyers, and peers to play opposing roles. As a result, many students graduate with limited hands-on exposure to the dynamics of live courtroom interaction before entering practice.

This project proposes an interactive, multi-agent AI system that simulates courtroom proceedings for law students. The system models the core participants of a trial — including a judge, prosecution, defense, and witness — as distinct AI agents that interact with each other and with the student in real time, allowing the student to practice legal reasoning and advocacy in a repeatable, low-stakes environment.

2. Objectives

1. Design and implement a multi-agent architecture that models the distinct roles and behaviors of courtroom participants.
2. Build an interactive simulation in which a law student can actively participate in a simulated trial stage (e.g., as counsel conducting or facing examination).
3. Ground the simulation's legal content in the Egyptian legal system, validated by a practicing legal professional.
4. Evaluate the system's pedagogical value and the realism of its simulated courtroom interactions.

3. Related Work & Novelty

Recent research has explored LLM-based agents for legal education and courtroom simulation. AgentsCourt and AgentCourt model multi-agent courtroom debate and judicial decision-making to support legal reasoning training. SimCourt simulates the broader judicial workflow from case presentation to deliberation, and a related 2026 study applied a multi-agent courtroom system to forensic-nursing testimony training within an interactive environment. Other work, such as LawThinker, focuses on legal research agents rather than courtroom interaction.

These systems are predominantly built around foreign legal systems (notably U.S. and Chinese procedure) or narrow professional-training contexts outside legal education itself. This project's contribution is a courtroom simulation grounded specifically in Egyptian legal procedure and legal Arabic, targeted at Egyptian law students — a combination not addressed in the existing literature.

4. Scope

To keep the project achievable within a two-semester timeline, the initial scope is deliberately narrow:

- A single case type (to be finalized with the legal domain consultant, e.g., a straightforward criminal misdemeanor case).
- A single trial stage per simulation session (e.g., witness examination or closing argument), rather than a full trial from start to end.
- A limited, fixed set of agent roles: judge, prosecution, defense, and witness — jury and multi-stage procedures are excluded from the initial scope.

Scope may be extended in later phases if development progresses ahead of schedule.

5. Data & Language Strategy

5.1 Data Sources

No public dataset of real Egyptian court transcripts exists, and such records are legally protected. The project will therefore rely on two data sources instead of transcript data:

- Retrieval-augmented generation (RAG) over the text of Egyptian procedural law (e.g., the Criminal/Civil Procedure Code), giving agents a grounded legal reference to draw on during interaction.
- A small set of synthetic case scenarios (3–5 initially), drafted and reviewed by the project's legal domain consultant to ensure procedural accuracy.

No fine-tuning of a language model on legal transcripts is planned, since sufficient training data does not exist; the system will rely on a general-purpose LLM combined with RAG grounding.

5.2 Language

The system must reflect the linguistic reality of Egyptian courtrooms, which mixes registers: formal legal Arabic (الفصحى القانونية) for the judge and attorneys, and colloquial Egyptian Arabic for lay witnesses. Producing accurate formal legal phrasing is a known limitation of general-purpose LLMs and will require close review by the legal domain consultant during Phase 1 and testing during Phase 5.

6. System Architecture (Overview)

The system will use a multi-agent orchestration framework (candidates: LangGraph, CrewAI) in which a controlling process manages turn-taking between agents based on the current stage of the simulated proceeding. Each agent (judge, prosecution, defense, witness) will be governed by a role-specific system prompt encoding its responsibilities, permitted actions, and procedural constraints, with access to the RAG legal knowledge base. The student interacts with the simulation through a dedicated interface, participating directly in the flow of the session rather than only observing it.

7. Learning Mechanism

Direct participation alone does not guarantee learning; the system is designed around a three-step cycle that turns each simulated interaction into a learning opportunity:

1. Participate — the student takes an active role (e.g., defense counsel) and makes real decisions within the simulated proceeding: framing questions, raising objections, or presenting arguments to the other agents.
2. Feedback — following each turn or at the end of the session, a dedicated evaluator agent (distinct from the role-playing agents) reviews the student's actions and provides structured feedback: whether a question was procedurally permissible, whether a legal argument was sound, and what a stronger alternative would have looked like.
3. Retry & Variation — the student can replay the same scenario with different decisions and observe how the simulated courtroom responds differently, reinforcing understanding through comparison rather than a single pass.

This participate-feedback-retry cycle is what distinguishes the system from a passive recording or scripted demonstration: the student's understanding is built through guided reflection on their own performance, not just exposure to a simulated scenario.

8. Project Timeline

Phase	Focus	Timeframe	Key Output
1. Research & Scoping	Literature review of multi-agent legal simulation systems; define target case type and trial stage; initial consultation with legal domain consultant	Semester 1 – Month 1	Scoped problem statement, related-work summary
2. System Design	Agent roles and orchestration logic; interaction design for the student user; technology stack selection	Semester 1 – Month 2	Architecture document
3. Initial Prototype	End-to-end working simulation for a single scenario (e.g., witness cross-examination)	Semester 1 – Month 3	Working demo, proposal defense
4. Full Development	Expand scenario coverage, refine agent prompts and behavior constraints, build student-facing interface	Semester 2 – Months 1–2	Functional system
5. Testing & Validation	Legal-accuracy review with domain consultant; usability testing with law students; iterative refinement	Semester 2 – Month 3	Validated system, evaluation results
6. Documentation & Defense Prep	Final report writing, presentation preparation, final polish	Semester 2 – Month 4	Final report, defense presentation

9. Evaluation Plan

The system will be evaluated along two dimensions:

- Legal accuracy and procedural realism, assessed through structured review by the project's legal domain consultant.
- Pedagogical value, assessed by comparing law students' understanding of the simulated procedure before and after using the system, where feasible.

10. Ethical Considerations

This system is an educational simulation tool intended to support law students' understanding of courtroom procedure. It is not a legal advice tool and is not a substitute for the judgment of a licensed legal professional. This limitation will be stated explicitly within the system and in all project documentation.

11. References

He, H. et al. AgentsCourt: Building Judicial Decision-Making Agents with Court Debate Simulation and Legal Knowledge Augmentation.

Chen, Z. et al. AgentCourt: Simulating Court with Adversarial Evolvable Lawyer Agents.

Godbole, M. et al. (2026). Multi-agent Courtroom Simulation System for Forensic Nursing Testimony Training. In: Artificial Intelligence in HCI, HCII 2026.

LLM Agents for Education: Advances and Applications (survey), Section A.3.3 Law Education.